$$E_3 = \left(x : 3\right) :: \left(f : fun_y \to e\right) :: \left(x : 1\right)$$

$$e = let_x = y + 1_in_\left(fun_z \to x + y + z\right)$$

$$\dfrac{\dfrac{\square}{E_3 \vdash f \Rightarrow fun_y \to e}\, Var____ \quad \dfrac{\square}{4 \Rightarrow 4}\, Constant____ \quad \dfrac{\dfrac{\square}{\left(y : 4\right) :: E_3 \vdash y + 1 \Rightarrow 5}\, Plus_____ \quad \dfrac{\square}{\left(x : 5\right) :: \left(y : 4\right) :: E_3 \vdash \left(fun_z \to x + y + z\right) \Rightarrow fun_z \to x + y + z}\, Constant}{\left(y : 4\right) :: E_3 \vdash e \Rightarrow fun_z \to x + y + z}\, Let}{E_3 \vdash \left(f_4\right) \Rightarrow fun_z \to x + y + z}\, App$$